

Improved drug–target interaction prediction with intermolecular graph transformer

Siyuan Liu[†], Yusong Wang[†], Yifan Deng, Liang He, Bin Shao, Jian Yin, Nanning Zheng, Tie-Yan Liu and Tong Wang

Corresponding author: Tong Wang, Tel: +86 10 5917 5872; E-mail: watong@microsoft.com

[†]Siyuan Liu and Yusong Wang contributed equally to this work.

Abstract

The identification of active binding drugs for target proteins (referred to as drug–target interaction prediction) is the key challenge in virtual screening, which plays an essential role in drug discovery. Although recent deep learning-based approaches achieve better performance than molecular docking, existing models often neglect topological or spatial of intermolecular information, hindering prediction performance. We recognize this problem and propose a novel approach called the Intermolecular Graph Transformer (IGT) that employs a dedicated attention mechanism to model intermolecular information with a three-way Transformer-based architecture. IGT outperforms state-of-the-art (SoTA) approaches by 9.1% and 20.5% over the second best option for binding activity and binding pose prediction, respectively, and exhibits superior generalization ability to unseen receptor proteins than SoTA approaches. Furthermore, IGT exhibits promising drug screening ability against severe acute respiratory syndrome coronavirus 2 by identifying 83.1% active drugs that have been validated by wet-lab experiments with near-native predicted binding poses. Source code and datasets are available at <https://github.com/microsoft/IGT-Intermolecular-Graph-Transformer>.

Keywords: drug–target Interaction, Intermolecular Graph Transformer, deep learning, drug discovery

Introduction

Human beings have always been suffering from various diseases. When a new disease such as coronavirus disease 2019 (COVID-19) breaks out, *de novo* drug design is not always the best option due to the huge cost in terms of both expense and time [1–3]. A possible alternative to this, which could allow to accelerate the process, is to perform drug repurposing, i.e. drug screening from known drugs and molecules [4–7].

With the rapid development of computational power, *in silico* drug screening that predicts the drug–target interactions (DTI) to identify active binding drug candidates has become one of the most important techniques in drug discovery [1, 2, 8]. Among the DTI prediction techniques, molecular/quantum dynamics simulations analyze the movements of atoms of proteins and drug molecules by applying physical or quantum mechanics and thus achieve high accuracy [9–11]. However, applying such simulations to high-throughput screening of

compound libraries is prohibitively expensive [2, 5, 9, 11]. Therefore, an alternative approach called molecular docking is widely adopted to trade accuracy for higher throughput by applying heuristics and empirical scoring functions [2, 11]. Molecular docking achieves a much higher throughput, but with the cost of a lower accuracy compared with the molecular/quantum dynamics approaches [2, 11].

Recently, many deep neural network models [12–17] have been proposed for DTI prediction, enabling high accuracy DTI prediction at affordable costs. These algorithms divide DTI prediction into two binary classification tasks, namely (i) binding activity prediction, which predicts whether a chemical compound can bind to the target protein and (ii) binding pose prediction, which determines whether the predicted binding pose is close to the native conformation. These deep neural network models can be roughly divided into two types. Given a compound–receptor pair, the first type of models (termed

Siyuan Liu is a PhD candidate at Sun Yat-sen University. He works as an intern at Microsoft Research Asia.

Yusong Wang is a PhD candidate at Xi'an Jiaotong University. He works as an intern at Microsoft Research Asia.

Yifan Deng is a graduate student at Fudan University. He works as an intern at Microsoft Research Asia.

Liang He is a senior researcher at Microsoft Research Asia. He received Ph.D. degree from University of Science and Technology of China. His research interests include computational biology and biological sequence modeling.

Bin Shao is a principal researcher at Microsoft Research Asia. He received his Ph.D. degree at Fudan University, Beijing, China. His current research interests include computational biology and bioinformatics.

Jian Yin is a professor at the data science and computer school of Sun Yat-sen University. His research interests include data mining and database.

Nanning Zheng is a Distinguished Professor of Xi'an Jiaotong University. His major research interests include computer vision, advanced intelligent computing architecture and autonomous vehicle.

Tie-Yan Liu is ACM/IEEE fellow and a Distinguished Scientist of Microsoft. He received his Ph.D. degree at Tsinghua University, Beijing, China. His current research interests include machine learning and AI for science.

Tong Wang is a senior researcher of Microsoft Research Asia. He received his Ph.D. degree at Tsinghua University, Beijing, China. His current research interests include computer-aided drug design, molecular dynamics simulation, bioinformatics and computational biology.

Received: January 10, 2022. **Revised:** March 28, 2022. **Accepted:** April 11, 2022

© The Author(s) 2022. Published by Oxford University Press. All rights reserved. For Permissions, please email: journals.permissions@oup.com

as the *type-I* approaches) [14, 16, 17] first dock the ligand in question to the receptor using molecular docking software, then extract the structure around the binding site from the resulting pose and finally feed the structure into the neural network. For example, Lim et al. [14] represent the binding site as a graph, and uses a graph neural network (GNN) to predict the binding activity and the binding pose. Specifically, this model adopts a distance-aware attention to handle intermolecular information, which greatly improves its performance on the DUD-E benchmark dataset [18]. The second type of model (termed as the *type-II* approaches) [12, 13, 15] has broader usage for receptors without 3D structures. In contrast with the *type-I* approaches, they first represent and process the receptor and ligand individually, and then combine the respective representations for binding activity prediction. For example, MONN [12] uses a CNN to encode full-length protein sequences to represent receptors and a GNN to represent the ligand, and the inner product of the two representations is used for the prediction.

These two types of models can be seen as two different yet complementary ways to model interactions between receptors and ligands, which sits at the center of the DTI problem. *Type-I* approaches leverage intermolecular information (i.e. coordinates and distances) from poses generated by molecular docking software. This information, albeit possibly inaccurate, reflects the physicochemical knowledge embodied in handcrafted scoring functions. However, these approaches usually treat the intermolecular edges indifferently as intramolecular edges, which in fact neglects the *topology* information (i.e. which edges are intermolecular). *Type-II* approaches choose not to use the generated poses, but instead learn the protein–ligand interactions with dedicated network modules. In this way, the prediction of binding activity is constrained to be a function of the learned interaction rather than the individual receptor and ligand, which also reflects the human inductive bias of the problem. Nonetheless, these models effectively only use the *topology* aspects of the intermolecular interactions. In summary, in both these two types of models, certain information about the intermolecular interactions is neglected and left to be implicitly learned by the neural network.

In view of this, we argue that the receptor–ligand interactions can be better modeled using a combined approach. We validate this argument by proposing and experimenting with a novel deep learning model called Intermolecular Graph Transformer (IGT). IGT takes the same inputs as the *type-I* approaches, while bearing an architectural resemblance to *type-II* approaches. In particular, the IGT has a three-way Transformer-based architecture [19, 20]. In each network block of the IGT, three graphs, namely the receptor graph, the ligand graph and the complex graph, are first individually processed and then combined through a specially designed mechanism, referred to as intermolecular attention. We show that this three-way design, as well as the intermolecular

attention, can greatly improve IGT's ability to learn the receptor–ligand interactions through an ablation study comparing the IGT with a one-way graph transformer counterpart.

We apply IGT to both binding activity prediction and binding pose prediction. For binding activity prediction, IGT is trained on two datasets, namely DUD-E and LIT-PCBA [21], respectively, and then evaluated on the test datasets. IGT surpasses the state-of-the-art (SoTA) models on almost all metrics, with 9.1% and 8.7% improvements on AUPRC, respectively, for DUD-E and LIT-PCBA. These performance gains demonstrate that IGT is able to capture the intermolecular information instead of simply fitting to the training samples. A notable finding from our experiments is that training on unbiased samples makes the models better for generalization. When evaluated on the independent test dataset MUV [22], a 5.3% increase of AUROC is witnessed, reflecting the better generalization capability of IGT. For pose prediction, IGT is trained and evaluated on PDBbind [23], which drastically outperforms molecular docking and achieves a 20.5% relative improvement to the second best. Finally, to test IGT on real-world DTI applications, we apply both the activity prediction model and the pose prediction model for drug virtual screening against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). When evaluated on Diamond SARS-CoV-2 drug dataset, IGT successfully identified the active drugs with near-native binding poses. Furthermore, IGT can serve as a general modeling framework for other fields regarding two-body interactions, such as protein–protein interaction. We sincerely hope IGT could accelerate practical drug discovery applications and inspire new ideas in future research of related fields.

The remainder of this manuscript is organized as follows. ‘Methods’ section describes the architecture of IGT, datasets and evaluation metrics. ‘Results’ section comprehensively demonstrates the performance of IGT for binding activity prediction and binding pose prediction, analyzes contributions of novel designed operations to performance gains by an ablation study and then applies IGT for real-world DTI scenarios. Finally, ‘Discussion and Conclusion’ section discusses advantages and limitations of this work and prospect future study for DTI prediction and drug virtual screening.

Methods

Intermolecular Graph Transformer

IGT is a novel graph transformer neural network for DTI based on the well-known Transformer architecture [19] and its generalization, Graph Transformer [20]. The original Transformer is currently the dominant model in natural language processing [19] and it has been shown transferable to and successful in other important fields, such as computer vision [24]. We adopt a variant of the dot-product attention from the Graph Transformer and apply it simultaneously to three graphs, i.e. the ligand graph, the receptor graph and the graph for the complex

structure. In addition, the graph dot-product attention layers are interleaved with an *intermolecular attention*, a dedicated graph-level operator we designed to better exploit the intermolecular edges in the graphs.

IGT consists of three modules: a feature extraction module, a message passing module and a readout module (Figure 1a). The feature extraction module extracts the physicochemical properties such as atom symbols, degrees, bond types as the raw atom and bond features. Furthermore, to distinguish the difference between the ligand and the receptor, we introduce a one-hot encoding of amino acid type and a three-dimensional residue identifier for each atom and bond of the receptor, respectively (Table S2 in supplementary materials). We concatenate all raw features to form the input vector α_i of atom i and bond input vector β_{ij} between atom i and atom j , and forms the hidden representation h_i^0, e_{ij}^0 with the positional encoding of molecular graphs λ_i (Eq. 1 and Eq. 2).

$$h_i^0 = (W_\alpha^0 \alpha_i + b_\alpha^0) + (W_\lambda^0 \lambda_i + b_\lambda^0) \quad (1)$$

$$e_{ij}^0 = W_\beta^0 \beta_{ij} + b_\beta^0 \quad (2)$$

where $W_\alpha^0 \in \mathbf{R}^{d \times d_n}$, $W_\beta^0 \in \mathbf{R}^{d \times d_e}$, $W_\lambda^0 \in \mathbf{R}^{d \times k}$ are learnable weight matrices, and $b_\alpha^0, b_\beta^0, b_\lambda^0 \in \mathbf{R}^d$ are learnable bias vectors. The hidden representations are then fed into the corresponding graphs in the message passing module, which consists of tandem repeated IGT blocks. In each IGT block, node and edge features of each graph are updated by a modified multi-head self-attention (MHA) mechanism (Eq. 3).

$$\tilde{h}_i^{l+1}, \tilde{e}_{ij}^{l+1} = \text{MHA}(h_i^l, e_{ij}^l) \quad (3)$$

where h_i^l is the feature of the node i in the layer l , e_{ij}^l is the feature of the edge between the node i and the node j , and $\tilde{h}_i^{l+1}$ is the hidden representation of the node i in the layer $l+1$ updated by MHA in the layer l . More specifically, before the softmax operation, the implicit attention information obtained from the multiplication of node *query* and key feature projections is further dot producted with the edge feature projections to obtain the modified attention information. The attention weights are then obtained by softmax operation, and the node features are updated by node *value* projections. At the same time, the implicit attention information is also used to update the edge features (Figure 1b). Then, we adopt an intermolecular attention (InterMA) to aggregate all information to update the complex graph (Eq. 4 and Figure 1c).

$$h_{com}^{l+1} = \text{InterMA}(\tilde{h}_{com}^{l+1}, \tilde{h}_{rec}^{l+1}, \tilde{h}_{lig}^{l+1}) \quad (4)$$

where the subscripts *com*, *rec* and *lig* represent the complex, receptor and ligand graph, respectively. The node features of complex graph in the final block are then fed into the readout module. All messages are aggregated by the aggregation operation and the binary score of binding activity or the binary score of binding

pose is predicted. The input representation of molecules, details of model architecture and training are described in the supplementary material.

Dataset preparation

We use the DUD-E [18] dataset and the LIT-PCBA [21] dataset to train, validate and test our model for binding activity prediction. DUD-E is the most widely used benchmark for binding activity prediction and the evaluation metric AUROC is reported on DUD-E by nearly all SoTA algorithms including MOON [12], DrugVQA [13], GNN-DTI [14], AtomNet [17] and so on. To first make an comparison with all SoTA algorithms, we trained and evaluated our model on DUD-E. It contains 22 886 active compounds against 102 targets and 50 decoys (i.e. inactive compounds) are generated for each active compound. After excluding the proteins that RDKit failed to load in the DUD-E dataset, we randomly divide the remaining 101 proteins into three disjoint sets, i.e. 71 proteins for training, 15 proteins for validation and 15 proteins for testing. Notably, latest studies [21, 25, 26] reported DUD-E may have bias since active compounds have distinct different physicochemical properties compared with inactive compounds. Thus, the algorithms tend to learn from ligands themselves instead of ligand–receptor pairs, resulting in poor generalization ability to unseen receptors and compounds.

To improve the generalization ability, we also trained and evaluated our model on LIT-PCBA dataset, a newly built binary labeled binding activity dataset that active and inactive compounds have similar properties. It contains 15 targets, 9780 active compounds and 407 839 inactive compounds and the original work of LIT-PCBA split the dataset into training and validation sets [21]. Following the original splitting scheme, we keep the training set and further evenly divide the original validation set into a new validation set and a test set. The dataset statistics are summarized in Table S1 in the supplementary material.

To further examine the generalization capability of our model, the MUV [22] dataset is used for evaluation. Since the MUV dataset does not provide 3D structures of the protein receptors, we retrieve protein structures for receptors from the PDB and discard the data with no PDB IDs, resulting in a total of 15 receptor proteins. Each receptor has about 30 active binding ligands and 15 000 inactive binding decoys. We generate the docked poses by Smina [27], which is a user-friendly version of Vina [28]. For active binding prediction, all samples in DUD-E, LIT-PCBA and MUV are docked with the default settings, and the conformations with the highest binding affinities calculated by Smina are picked up for model training and evaluation.

For binding pose prediction, we adopt PDBbind refined set [23] and split it into training, validation and test sets with a ratio of 0.70:0.15:0.15. We also used Smina to generate poses for each receptor–ligand pair. All docked poses are aligned to the corresponding crystal structures.

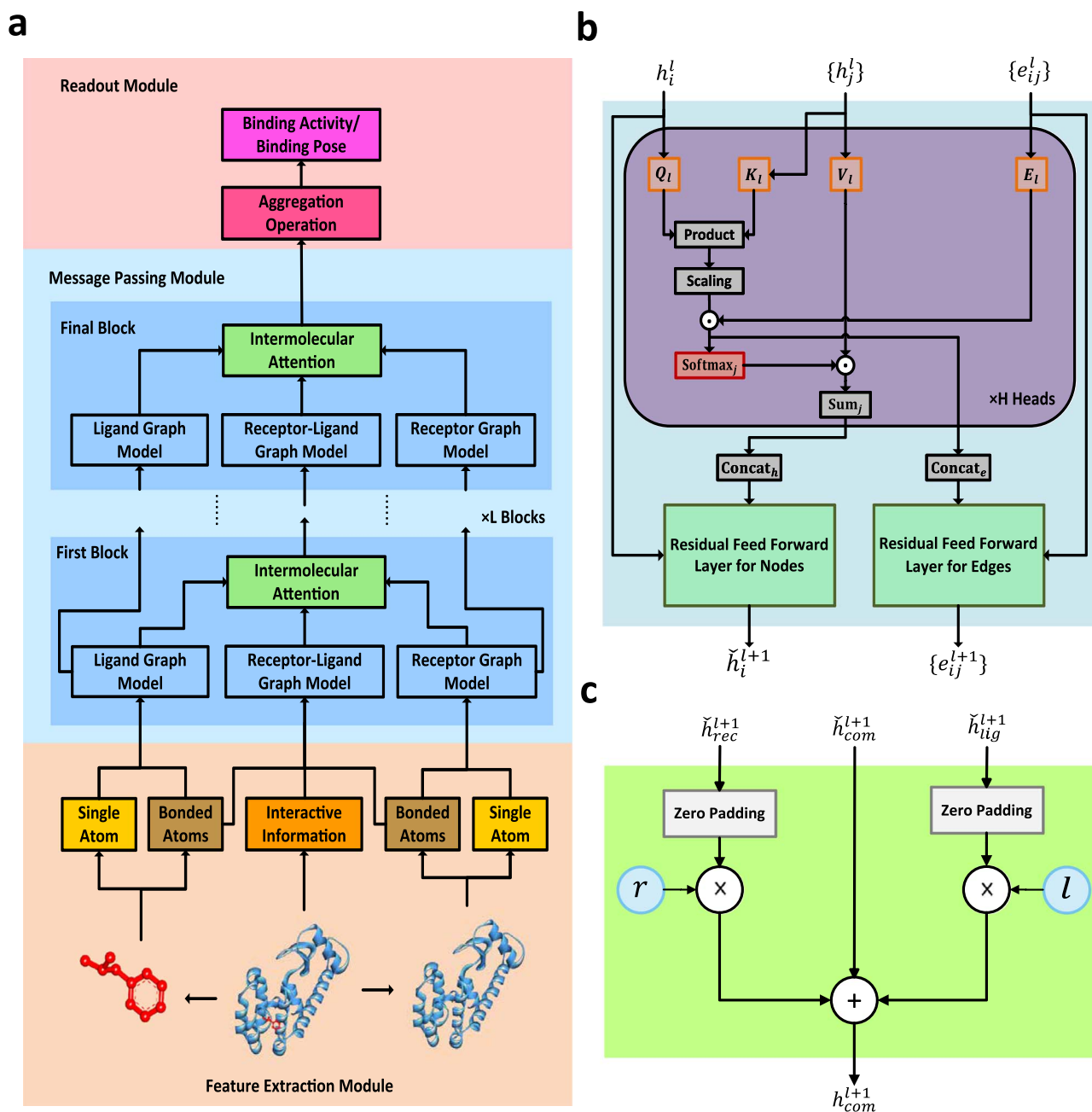


Figure 1. The overall model architecture of IGT, where h_i^l is the feature of the node i in the layer l , e_{ij}^l is the feature of the edge between the node i and the node j , $\tilde{h}_i^{l+1}$ is the hidden representation of the node i in the layer $l+1$ updated by multi-head attention. $\tilde{h}_{com}^{l+1}$ is the hidden representation of the complex graph after intermolecular attention. Q_l , K_l , V_l , E_l are learnable weight matrices, and r , l are learnable weight scalars. **(A)** The flowchart of IGT. The feature extraction module, the message passing module and the readout module are shown in the orange, blue and pink panels, respectively. **(B)** The architecture of graph model in the message passing module. Node features as well as positional encodings pass through a multi-head attention module, while the edge features act as weights for each attention between two nodes. Then node features are updated by residual feed forward layers. **(C)** The depiction of intermolecular attention. Attention weights are assigned to the ligand, receptor and complex graphs. The node features of the complex graph are updated by the information of all the three graphs.

The poses with *root-mean-square deviation* (RMSD) values less than 2.0 Å are labeled as positive samples, those with RMSD greater than 4.0 Å are labeled as negative samples and the remaining are discarded. Since we split the dataset by receptors and labeled the samples by poses, the numbers of samples in the validation set and independent test set are different though they consist of the same number of receptors.

Furthermore, to evaluate IGT's generalization capability to novel receptors, we adopt the Diamond SARS-CoV-2 drug dataset (<https://www.diamond.ac.uk/covid-19/for-scientists/Main-protease-structure-and-XChem.html>) that consists of 879 active and inactive drugs against SARS-CoV-2 main protease derived from cell and structural experiments. Smina is run for each drug with default parameters and the docked poses with the lowest

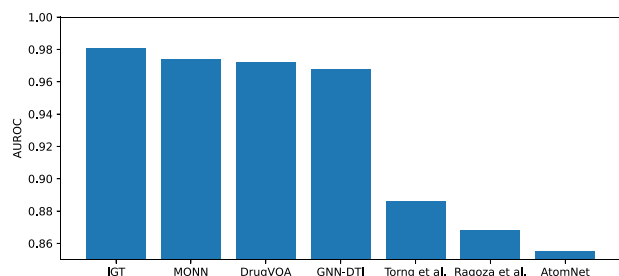


Figure 2. Comparison of AUROC on DUD-E, among IGT and other approaches [12–17]. For the convenience of comparison, the AUROC on y-axis is shown from 0.85.

energy values are picked up for active binding prediction. For the drugs predicted as active compounds, we further conduct pose predictions among all docked poses.

Evaluation

We select seven evaluation metrics for both active binding prediction and pose prediction, i.e. AUROC, LogAUC, AUPRC, ROC enrichment, enrichment factor, balanced accuracy and Matthews correlation coefficient (MCC) (see supplementary materials for more details).

Results

Performance on binding activity prediction

To evaluate the performance of IGT for binding activity prediction, we first use the widely adopted DUD-E dataset as a benchmark. IGT is compared with two types of deep learning-based DTI prediction approaches as well as molecular docking. As demonstrated before, type-I approaches including AtomNet [17], Ragoza et al. [16] and GNN-DTI [14] take the complex structures after docking as inputs and learn from the whole docked structures, while type-II approaches including Tornø et al. [15], DrugVQA [13] and MONN [12] learn from ligand and receptor, respectively, without the docked complex structures. Inspired by the advantages of both types of approach, IGT makes full use of the docked poses by learning the complex structure and the respective drug and protein simultaneously. As demonstrated in ‘Data preparation’ section, we split DUD-E into training, validation, and test sets. The IGT is trained on the training and validation sets with a learning rate of $1e-5$ (see supplementary material for more details), and then evaluated on the test set. As shown in Figure 2, IGT achieves a remarkable AUROC of 0.981 on DUD-E and outperforms all type-I and type-II approaches. Compared with the poor performance of the *vanilla* molecular docking method, all the deep learning-based approaches achieved much higher AUROC.

To eliminate the effect of data splitting schemes, we further select the SoTA type-II approach MONN [12] and the SoTA type-I approach [14] (referred to as GNN-DTI in the following text for convenience) for reproduction with the same data splitting scheme during model training

and evaluation. We also evaluate performance on several metrics besides AUROC, i.e. AUPRC, adjusted LogAUC, ROC enrichment, balanced accuracy, enrichment factor and MCC to avoid the potential bias of a single metric. As shown in Table 1, IGT achieves the best performance in terms of all seven metrics, which shows its superior capability for DTI prediction.

We also employ LIT-PCBA for model training and evaluation. Following the same procedures for DUD-E, we also split LIT-PCBA into training, validation and test sets, and train IGT with the identical hyper-parameters. MONN and GNN-DTI are also trained on the same dataset with their default hyper-parameters. As shown in Table 1, we compare the performance of IGT with that of MONN and GNN-DTI. The *vanilla* molecular docking method only achieved a 0.558 AUROC score on LIT-PCBA, significantly lower than that on DUD-E and the other three approaches on LIT-PCBA. Our IGT model performs the best in terms of six metrics out of the seven.

Generalization ability to unseen receptors and compounds

IGT has achieved superior performance on both DUD-E and LIT-PCBA. However, those results do not necessarily reflect the performance of the models on real DTI applications since unseen receptors and/or compounds are often involved in real-world scenarios, such as virtual drug screening for novel targets. Therefore, we further evaluate the models on MUV, a new test dataset that consists of different receptor proteins from DUD-E and LIT-PCBA. As demonstrated above, we train IGT and reproduce MONN and GNN-DTI with their default hyper-parameters on DUD-E and LIT-PCBA, respectively, resulting in a total of six models for binding activity prediction. Meanwhile, the *vanilla* molecular docking is also treated as a baseline model. All these models were evaluated on the MUV test set, and their performance is compared in Table 2 and Figure S1.

Two observations can be drawn from these results. First, for all these three approaches, the performance of a prediction model trained on LIT-PCBA is better than that of the same model trained on DUD-E, indicating that training on unbiased samples does increase model performance on unseen receptors and compounds. Second, although performance gains are observed in all the three approaches when trained on LIT-PCBA, the improvement for MONN is only 0.014, which is relatively small compared with 0.087 for IGT and 0.062 for GNN-DTI. This implies that even if a DTI model is trained on unbiased data, its generalization capability still heavily depends on model design.

This result shows that IGT is a competitive solution for real-world DTI applications because of its superior generalization ability. However, one might question what leads to IGT’s better performance. To answer this question, we conduct ablation studies as reported in the next section.

Table 1. Performance evaluation of binding activity prediction on DUD-E and LIT-PCBA.

Dataset	Model	AUROC	LogAUC	AUPRC	Bal. Acc.	ROC Enrich.	Enrich. Factor	MCC
DUD-E	IGT	0.981	0.754	0.730	0.875	75.9	36.6	0.638
	MONN	0.969	0.705	0.669	0.768	66.2	32.8	0.625
	GNN-DTI	0.960	0.668	0.581	0.864	58.7	29.7	0.409
	Docking	0.707	0.156	0.139	–	10.4	6.35	–
LIT-PCBA	IGT	0.942	0.586	0.311	0.871	40.1	23.3	0.315
	MONN	0.940	0.582	0.286	0.861	38.6	23.4	0.305
	GNN-DTI	0.925	0.549	0.253	0.839	32.2	20.4	0.308
	Docking	0.558	0.035	0.013	–	2.36	1.83	–

Table 2. Evaluation of the generalization ability among IGT, MOON, GNN-DTI and Docking on MUV test set.

Model	AUROC (trained on DUD-E)	AUROC (trained on LIT-PCBA)
IGT	0.547	0.634
MONN	0.546	0.560
GNN-DTI	0.540	0.602
Docking	0.547 (No training required)	

Ablation study

To better understand where the performance gains come from, we conduct an extensive ablation study on both the features used in IGT and the model architecture we design. Different sets of features or model components are removed from the models, and the resulting models are trained on LIT-PCBA and evaluated on the validation set with AUROC and AUPRC.

Features. As shown in Table S4, we first remove all node features except the basic atom symbol, resulting in a 0.030 and 0.097 decrease of the AUROC and AUPRC score, respectively. This result indicates that the atom features do contribute significantly to model performance as expected. We then remove the feature of intermolecular distances, which also leads to a large performance drop. In contrast, removing all edge features but the intermolecular distances resulted in only a small decrease of AUROC and AUPRC score. This indicates that the inter-atomic distances are probably more important than other edge features. Furthermore, performance drops are observed in all three experiments, showing that both atom and edge features contribute to the DTI predictions. The ablation study on features shows that the IGT can learn from both the docked structure (e.g. the distance feature) and the physicochemical properties of the individual atoms.

Modules. We further conduct the ablation study on the neural network architecture of the model. We devise two experiments by modifying the model architecture to study how the model learns intermolecular information. When the intermolecular attentions are removed (i.e. resulting in a one-way graph transformer), a 0.016 AUROC and 0.064 AUPRC performance drop is observed. For the complex graph in each building block, removing intramolecular edge features results in a similar performance drop. This result demonstrates that both the

modeling of the separated components and that of their interactions are important to the DTI prediction.

Three-way versus one-way. We further investigate the effect of the intermolecular attention and the three-way design. To study whether such design can truly improve the model's ability to capture intermolecular interactions, we compare the IGT with its one-way counterpart, using the unsupervised dot-product attention scores. The unsupervised dot-product attention scores are calculated as in Eq. 1.

$$A = \text{softmax}(HH^T) \quad (5)$$

where $H \in \mathbf{R}^{N \times d}$ denotes the graph node embedding tensor. The entry A_{ij} can be seen as the contribution of node j to node i . We calculate the attention scores A right before the final prediction layers for both the three-way and one-way models, and then compute the mean of the entries of A that correspond to intermolecular interactions (i.e. A_{ij} such that i belongs to ligand and j belongs to receptor, and vice versa). This score, which we call the mean intermolecular attention score, reflects how much the intermolecular interactions contribute to node representations. The procedure is repeated for all proteins in the test set to obtain a distribution. As shown in Figure 3a, the distributions for the two models are significantly different. Concretely, the distribution for the three-way model is much wider and right-shifted than that of the one-way model, indicating that prediction in the three-way model relies more on intermolecular information. This confirms that the three-way design, as well as the intermolecular attention mechanism, greatly improves the modeling of intermolecular interactions, explaining the better performance of IGT.

Ligand-Only. We further investigate the effect of protein information by designing a ligand-only model. Only the ligand graph is fed into a one-way graph Transformer; a significant 0.206 and 0.267 decrease of AUROC and AUPRC score is observed. This observation demonstrates that the information embedded in the receptor and the receptor–ligand interactions is crucial to DTI prediction, and our three-way model design captures the information well.

DUD-E versus LIT-PCBA. As mentioned in previous sections, it is well acknowledged that DUD-E contains various types of biases and is not suitable to serve as

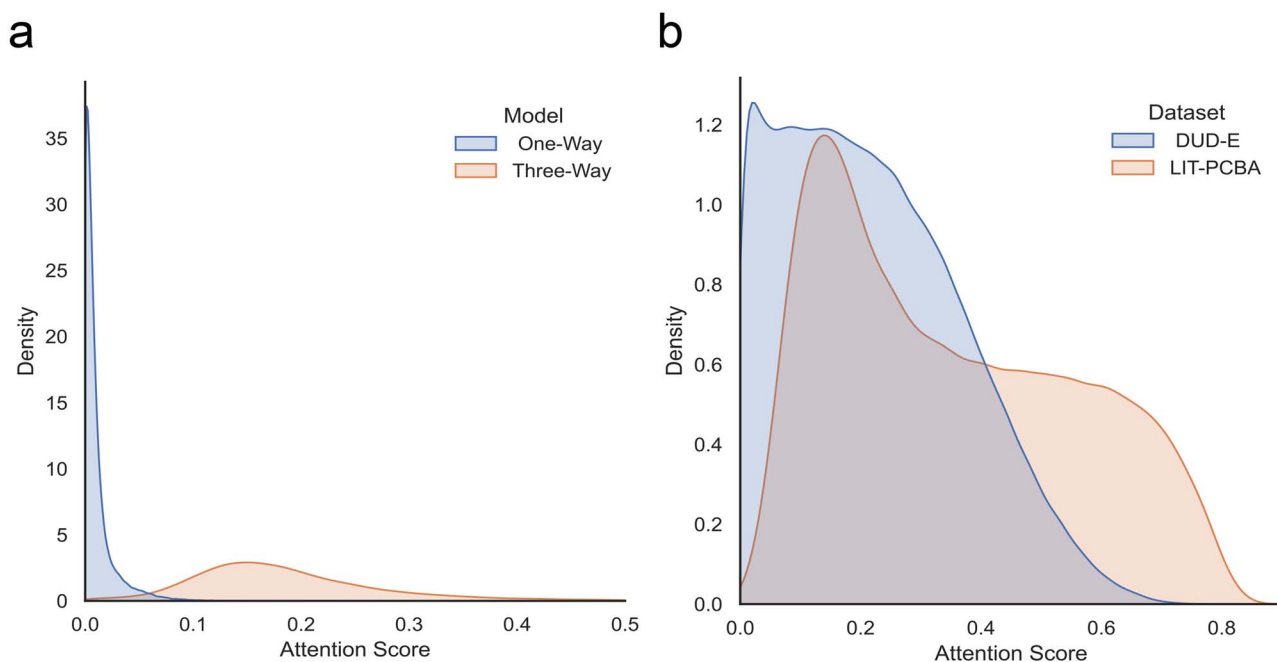


Figure 3. The distributions of mean intermolecular attention scores. **(A)** Comparison between the distribution of the three-way model and that of the one-way model. The two models are both trained and evaluated on LIT-PCBA training and test sets. **(B)** Comparison between the distribution of the three-way models on MUV test set. The two models are trained on DUD-E and LIT-PCBA datasets, respectively.

a training set for machine learning. In this section, we quantitatively analyze this problem by comparing the distributions of mean intermolecular attention scores for two IGT models, one trained on DUD-E and the other trained on LIT-PCBA. As shown in Figure 3b, the distribution for the LIT-PCBA model is more right-shifted than the one for the DUD-E model, meaning that the model trained on LIT-PCBA pays more attention to the intermolecular interactions. This provides an explanation of IGT's super generalization ability over previous models: the correct choice of an unbiased training dataset and the reasonable model design of IGT jointly allow for the successful learning of intermolecular information, which is the key to generalization.

Performance on binding pose prediction

Searching the binding pose for a given complex closest to the native conformation is essential to understanding intermolecular interactions. As described in the Dataset Preparation section, we adopt the identical model architecture as the one used for binding activity prediction and trained and evaluated IGT on PDBbind [23]. MONN is a type-II model and cannot be applied to this task, since all candidate poses share the same input representations. As shown in Table 3, the performance of both type-I models for binding pose prediction is much better than that of the vanilla molecular docking method. Compared with the best type-I approach GNN-DTI [14], the AUROC, AUPRC and MCC scores of our model are 0.061, 0.130 and 0.137 higher, respectively. As mentioned in binding activity prediction, our IGT model is capable of modeling the structures of the complexes and therefore improves the performance on binding pose prediction.

Table 3. Performance evaluation of binding pose prediction on PDBbind.

Model	AUROC	AUPRC	MCC
IGT	0.915	0.765	0.672
GNN-DTI	0.854	0.635	0.535
Docking	0.702	0.466	–

Figure S2 shows the percentages of poses whose RMSD are smaller than 2 Å within the top-K poses ranked by IGT, GNN-DTI and the docking method, respectively. Compared with the docking method and GNN-DTI, our model shows greater ability to identify near-native poses.

Application for real-world DTI scenarios

To further validate the practical value of IGT, the Diamond SARS-CoV-2 drug dataset is used as a real-world evaluation. We prepare two models of IGT for this task, i.e. the binding activity prediction model (M1) and the binding pose prediction model (M2). The M1 model is trained on LIT-PCBA, while M2 is trained on PDBbind. The molecular docking software, Smina, is run to generate binding poses. M1 is then used to predict the activity of ligands against the SARS-CoV-2 main protease. As a result, the IGT M1 model successfully recalls 83.12% active ligands and achieves an AUROC score of 0.701. Then we select the active binding ligands that are also predicted as the active ones by M1 model and identify the best binding pose for each such protein–ligand pair using the M2 model. As shown in Figure 4, the predicted binding pose selected by the M2 model is basically indistinguishable from the native structure with an RMSD of 1.456 Å, indicating that our IGT

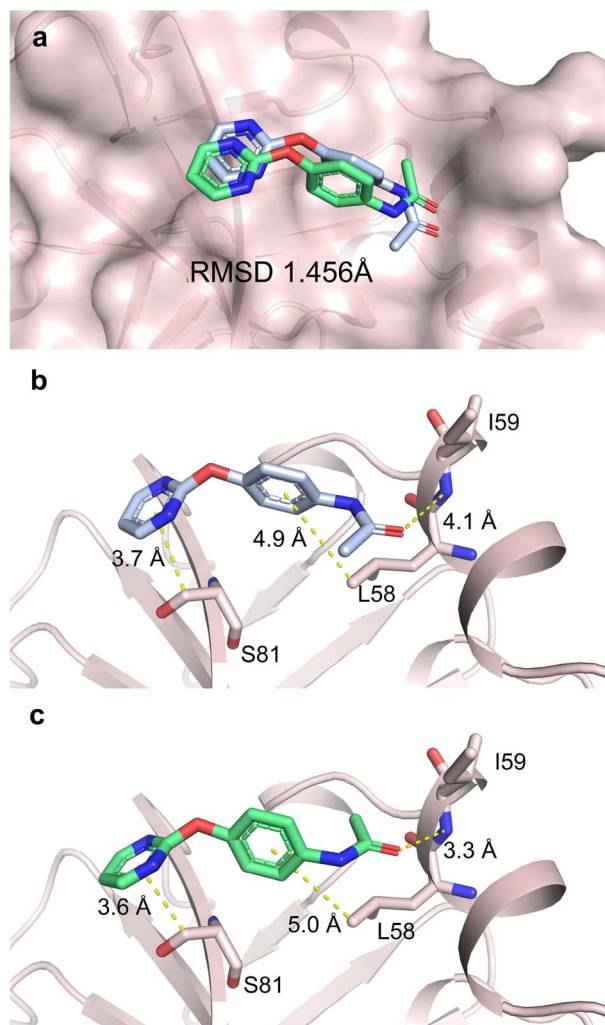


Figure 4. Case study for drug virtual screening against SARS-CoV-2 by IGT model. **(A)** The docked poses of the drug-protein complex. The predicted binding pose and the ground truth are colored green and lilac, respectively. The binding pocket of the receptor protein is colored amaranth with surface. The RMSD value between the predicted binding pose and the ground truth is 1.456 Å. **(B)** Interaction analysis between the ground truth ligand pose and the protein. **(C)** Interaction analysis between the predicted binding pose and the receptor. In **(B)** and **(C)**, the ligands as well as interactive residues are shown as sticks and the remaining protein structures are shown as secondary structures.

model successfully detects the near-native pose against SARS-CoV-2. The predicted binding pose forms similar interactions that can also be found in the native complex structure. In particular, in the predicted drug binding pose, a π - σ interaction and a hydrophobic interaction are formed between S81 and L58, respectively. Furthermore, a stronger hydrogen bond is formed with the residue I59 compared with that in the native pose, indicating a more stable binding to the pocket of the receptor protein.

The evaluation on the SARS-CoV-2 drug dataset shows that our model not only performs well on benchmark datasets like DUD-E, LIT-PCBA or MUV, but also has its practical value to be applied to real-world DTI scenarios.

Discussion and Conclusion

We have demonstrated the superior performance of IGT in terms of both fitting and generalization capabilities through the evaluations on the DUD-E, LIT-PCBA, MUV and the SARS-CoV-2 main protease DTI datasets. The factors that contribute to the superior performance of IGT can be summarized as follows. First, the architecture of IGT assigns more weights to intermolecular information than intramolecular information, so it can better reflect the physical rules in DTIs as well as avoid fitting to the biases in the data. Second, the design of IGT is more robust to the spatial relationship between the ligand and the receptor. For example, the predictions of IGT are invariant to translations and rotations of the inputs (*translational and rotational equivariance*) because it only leverages features invariant to such transformations. In contrast, this deserved property is seldom found in the voxelization-based methods (e.g. 3D CNNs). Third, training on the unbiased LIT-PCBA dataset allows IGT to fit to the signal rather than the biases in the data. Furthermore, IGT could be combined with other types of approaches for drug virtual screening. For example, the Type II approaches are employed as a large-scale virtual screening tool for all ligand-receptor pairs to significantly reduce the number of candidate ligands. Subsequently, molecular docking is adopted to make a coarse-grained screening and generate the docked poses, followed by IGT and Type I approaches to further refine docking results. The precise binding free energy could be estimated by molecular dynamics/quantum dynamics simulations.

In addition, for binding activity prediction, besides the task made by type-I, type-II approaches and ours that predicts whether a ligand can bind to the receptor or not, there exists another task that directly regress the real-valued binding affinities. Although models and algorithms for this task [29–33] employ obviously different datasets and model design, we reproduced the SoTA algorithms MDeePred and DeepDTA on DUD-E and LIT-PCBA datasets, respectively, by modifying the last layer of their models with a sigmoid function for a binary-valued prediction. As shown in Table S5 in the supplementary material, IGT outperformed MDeePred and DeepDTA with the highest AUROC and AUPRC values on DUD-E dataset. For LIT-PCBA, IGT also achieved the best performance on AUROC, while MDeePred had better performance on the other metric.

Although IGT performs well on both binding activity prediction and pose prediction, it still has a few limitations. First, IGT requires the docking poses from molecular docking software as input. This may lead to error propagation since the molecular docking procedure itself is nondeterministic and inaccurate. Second, molecular docking slows down the execution of IGT, hindering its efficiency of virtually screening active drugs from millions to billions chemical compounds. Third, although IGT exhibits superior generalization ability to unseen

receptors and compounds, the performance is still much lower than that evaluated on similar test sets. Recently, AlphaFold2 [34] has been proved to make a disruptive improvement for protein structure prediction. For receptors without resolved structures, predicted structures by AlphaFold2 is an alternative choice for IGT to extend its application for those receptors. Designing a DTI prediction model with better generalization ability, lower resource consumption and broader usage still awaits future study.

Key Points

- DTI prediction plays an essential role in *in silico* drug discovery and can be divided into active binding prediction and active pose prediction.
- We propose the model, IGT based on a three-way graph Transformer, and it achieves the best performance for two prediction tasks.
- We design intermolecular attention to both capture more interactive information for DTI and interpret prediction results.
- IGT shows better generalization ability to unseen receptors and can be used for anti-COVID-19.

Supplementary data

Supplementary data are available at *Briefings in Bioinformatics* online.

Author contributions statement

S.L. contributed to methodology, modeling, data analysis, software and writing the original draft. Y.W. contributed to modeling, data analysis and software and writing the original draft. T.W. contributed to experimental design, methodology, data analysis and writing the original draft. Y.D. contributed to data analysis. L.H. contributed to data analysis and writing. B.S. contributed to experimental design and writing. J.Y., N.Z. and T.L. contributed to writing. All authors reviewed the final manuscript.

Acknowledgments

The authors thank the reviewers for their valuable comments and suggestions.

Funding

This work has been supported by Microsoft Research Asia, the National Natural Science Foundation of China (U1711262, U1711261, U1811264, U1811261, U1911203, U2001211), Guangdong Basic and Applied Basic Research Foundation (2019B1515130001), and Key R&D Program of Guangdong Province (2018B010107005).

References

1. da Silva Rocha SFL, Olanda CG, Fokoue HH, et al. Virtual screening techniques in drug discovery: review and recent applications. *Curr Top Med Chem* 2019;**19**(19):1751–67.
2. Lin X, Li X, Lin X. A review on applications of computational methods in drug screening and design. *Molecules* 2020;**25**(6):1375.
3. Canrong W, Liu Y, Yang Y, et al. Analysis of therapeutic targets for sars-cov-2 and discovery of potential drugs by computational methods. *Acta Pharmaceutica Sinica B* 2020;**10**(5):766–88.
4. Ashburn TT, Thor KB. Drug repositioning: identifying and developing new uses for existing drugs. *Nat Rev Drug Discov* 2004;**3**(8):673–83.
5. Novac N. Challenges and opportunities of drug repositioning. *Trends Pharmacol Sci* 2013;**34**(5):267–72.
6. Xue H, Li J, Xie H, et al. Review of drug repositioning approaches and resources. *Int J Biol Sci* 2018;**14**(10):1232.
7. Dudley JT, Deshpande T, Butte AJ. Exploiting drug–disease relationships for computational drug repositioning. *Brief Bioinform* 2011;**12**(4):303–11.
8. Csermely P, Korcsmáros T, Kiss HJM, et al. Structure and dynamics of molecular networks: a novel paradigm of drug discovery: a comprehensive review. *Pharmacol Ther* 2013;**138**(3):333–408.
9. De Vivo M, Masetti M, Bottegoni G, et al. Role of molecular dynamics and related methods in drug discovery. *J Med Chem* 2016;**59**(9):4035–61.
10. Durrant JD, Andrew J, McCammon. Molecular dynamics simulations and drug discovery. *BMC Biol* 2011;**9**(1):1–9.
11. Śledź P, Caflisch A. Protein structure-based drug design: from docking to molecular dynamics. *Curr Opin Struct Biol* 2018;**48**:93–102.
12. Li S, Wan F, Shu H, et al. Monn: a multi-objective neural network for predicting compound–protein interactions and affinities. *Cell Systems* 2020;**10**(4):308–22.
13. SHUANGJIA Zheng, YONGJIAN Li, SHENG Chen, Jun Xu, and YUEDONG Yang. Predicting drug–protein interaction using quasi-visual questionanswering system. *Nature Machine Intelligence*, **2**(2):134–40, Feb 2020.
14. Lim J, Ryu S, Park K, et al. Predicting drug–target interaction using a novel graph neural network with 3d structure-embedded graph representation. *J Chem Inf Model* 2019;**59**(9):3981–8.
15. Torng W, Altman RB. Graph convolutional neural networks for predicting drug–target interactions. *J Chem Inf Model* 2019;**59**(10):4131–49.
16. Ragoza M, Hochuli J, Idrobo E, et al. Protein–ligand scoring with convolutional neural networks. *J Chem Inf Model* 2017;**57**(4):942–57.
17. Wallach I, Dzamba M, Heifets A. Atomnet: a deep convolutional neural network for bioactivity prediction in structure-based drug discovery. arXiv preprint arXiv:1510.02855. 2015.
18. Mysinger MM, Carchia M, Irwin JJ, et al. Directory of useful decoys, enhanced (dud-e): better ligands and decoys for better benchmarking. *J Med Chem* 2012;**55**(14):6582–94.
19. Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need[J]. *Advances in neural information processing systems*. 2017, 30.
20. Dwivedi VP, Bresson X. A generalization of transformer networks to graphs[J]. arXiv preprint arXiv:2012.09699. 2020.
21. Tran-Nguyen V-K, Jacquemard C, Rognan D. Lit-pcba: An unbiased data set for machine learning and virtual screening. *J Chem Inf Model* 2020;**60**(9):4263–73.
22. Rohrer SG, Baumann K. Maximum unbiased validation (muv) data sets for virtual screening based on pubchem bioactivity data. *J Chem Inf Model* 2009;**49**(2):169–84 PMID: 19161251.

23. Liu Z, Minyi S, Han L, et al. Forging the basis for developing protein–ligand interaction scoring functions. *Acc Chem Res* 2017;**50**(2):302–9.
24. Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929. 2020.
25. Xia J, Tilahun EL, Reid T-E, et al. Benchmarking methods and data sets for ligand enrichment assessment in virtual screening. *Methods* 2015;**71**:146–57.
26. Chen L, Cruz A, Ramsey S, et al. Hidden bias in the dud-e dataset leads to misleading performance of deep learning in structure-based virtual screening. *PloS one* 2019;**14**(8):e0220113.
27. Koes DR, Baumgartner MP, Camacho CJ. Lessons learned in empirical scoring with smina from the csar 2011 benchmarking exercise. *J Chem Inf Model* 2013;**53**(8):1893–904.
28. Trott O, Olson AJ. Autodock vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J Comput Chem* 2010;**31**(2): 455–61.
29. Öztürk H, Özgür A, Ozkirimli E. Deepdta: deep drug–target binding affinity prediction. *Bioinformatics* 2018;**34**(17):i821–9.
30. Karimi M, Di W, Wang Z, et al. Deepaffinity: interpretable deep learning of compound–protein affinity through unified recurrent and convolutional neural networks. *Bioinformatics* 2019;**35**(18):3329–38.
31. Rifaoglu AS, Atalay RC, Cansen Kahraman D, et al. Mdeepred: novel multi-channel protein featurization for deep learning-based binding affinity prediction in drug discovery. *Bioinformatics* 2021;**37**(5):693–704.
32. Cichonska A, Ravikumar B, Parri E, et al. Computational-experimental approach to drug-target interaction mapping: a case study on kinase inhibitors. *PLoS Comput Biol* 2017;**13**(8):e1005678.
33. He T, Heidemeyer M, Ban F, et al. Simboost: a read-across approach for predicting drug–target binding affinities using gradient boosting machines. *J Chem* 2017;**9**(1):1–14.
34. Jumper J, Evans R, Pritzel A, et al. Highly accurate protein structure prediction with alphafold. *Nature* 2021;**596**(7873):583–9.